TEST REPORT

Control No. : IL – QL – 21075

Period : 2020.12.01 ~ 2021.01.11

Purpose
Quality Check

SIGN	Prepared	Check	Review	Approval
	L.K.C	–	P.T.S	K.J.J
	(SIGNED)	–	(SIGNED)	(SIGNED)

Model	T 6.5, T 10	Test contact	Lee Ki–Cheol
Part no.	T6.5 : GRF5–514C5	Test method	PW67601 Drawing SPEC
Part no.	T10 : GRF5–55431		
Part name	LED BULB	Grade	–
Manufacturer	IL HEUNG CO., LTD.	Date	2020.12.01

[1] Purpose

◆ Reliance Test for LED supplier change. (SAMSUNG to SEOUL SEMI)

[2] Item

LEG	NO	TEST ITEM	RESULT	REMARKS
PW67601				
	1-1	Performance evaluation/Functional evaluation	**PASS**	
	1-2	Low Temperature Operation	**PASS**	
	1-3	High Temperature Operation	**PASS**	
	1-4	Thermal Cycle	**PASS**	
	1-5	Thermal Shock	**PASS**	
	1-6	Temperature Humidity Cycle	**PASS**	
	1-7	Performance evaluation/Functional evaluation	**PASS**	
Drawing Spec'				
2		Illuminnation	**PASS**	
Etc				
3		High temp & humidity actual vehicle test	**PASS**	

[3] Tester opinion

◆ Test result is satisfied according to the above issue test.

IL - QL - 20781	1-1. Performance evaluation/Functional evaluation, Initial
Test Date	2020.12.01
Test Sample	T 6.5 5ea, T 10 5ea

Performance & Functional evaluation for T 6.5

T 6.5	Functional Test					Test result
	Voltage Drop	Insulation Resistance	Current Check	There shall be no alfunction and abnormality		
SPECIFICATION (12V)	0.1V (MAX)	Shall be 1MΩ min.	After 30min aging.			
S1	13.4 mV	∞	90.7	PASS	PASS	PASS
S2	13.6 mV	∞	90.8	PASS	PASS	PASS
S3	13.8 mV	∞	90.3	PASS	PASS	PASS
S4	13.7 mV	∞	90.6	PASS	PASS	PASS
S5	13.3 mV	∞	91.2	PASS	PASS	PASS

T 6.5	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5216	0.3395	0.3504	PASS
S2	PASS	PASS	5220	0.3395	0.3517	PASS
S3	PASS	PASS	5203	0.3398	0.3502	PASS
S4	PASS	PASS	5209	0.3397	0.3512	PASS
S5	PASS	PASS	5135	0.3416	0.3512	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	1-1. Performance evaluation/Functional evaluation, Initial
Test Date	2020.12.01
Test Sample	T 6.5 5ea, T 10 5ea

Performance & Functional evaluation for T 10

T 10	Functional Test					Test result
	Voltage Drop	Insulation Resistance	Current Check	There shall be no alfunction and abnormality		
SPECIFICATION (12V)	0.1V (MAX)	Shall be 1MΩ min.	After 30min aging.			
S1	23.1 mV	∞	64.6	PASS	PASS	PASS
S2	23.4 mV	∞	64.8	PASS	PASS	PASS
S3	23.8 mV	∞	64.6	PASS	PASS	PASS
S4	23.1 mV	∞	64.7	PASS	PASS	PASS
S5	23.4 mV	∞	64.7	PASS	PASS	PASS

T 10	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5186	0.3403	0.3511	PASS
S2	PASS	PASS	5182	0.3404	0.3508	PASS
S3	PASS	PASS	5193	0.3401	0.3509	PASS
S4	PASS	PASS	5176	0.3406	0.3515	PASS
S5	PASS	PASS	5176	0.3406	0.3512	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	1-2. Low Temperature Operation
Test Date	2020.12.03 ~ 12.06
Test Sample	T 6.5 5ea, T 10 5ea

Test Standard

(1) Put the test sample into the thermostatic chamber at the standard temperature.
Gradually lower the ambient temperature to the minimum dependable operating temperature **(T1, -40℃)**, and leave it untouched at the temperature for 1 h or more. **Operate(12V)** it at the temperature for **72 h**.
(2) Check the test sample for malfunction during the operation.
(3) Take the test sample out of the chamber. If its surface is moist with water droplets, remove them and leave it untouched for 2 h or more under the standard condition.
(4) Evaluate the performance, function and visual inspection.

EUT Conditions

Functional states class
12 V, -40˚C, 72h

EUT Conditions

Test items	Parameters	Result	
		T 6.5	T 10
		5 Samples	5 Samples
Low Temperature Operation	- The appearance of parts and components shall not have any trace that may adversely affect performance and function. "Breakage, fractures, cracks, scraping, dimensional changes, corrosion, electrolytic corrosion, discoloration, water intrusion, etc." - As for solder cracking, the cracking rate shall be under 50 % when the largest cracking portion on the fillet is viewed from above. - No trace of smoking and firing.	**PASS**	**PASS**

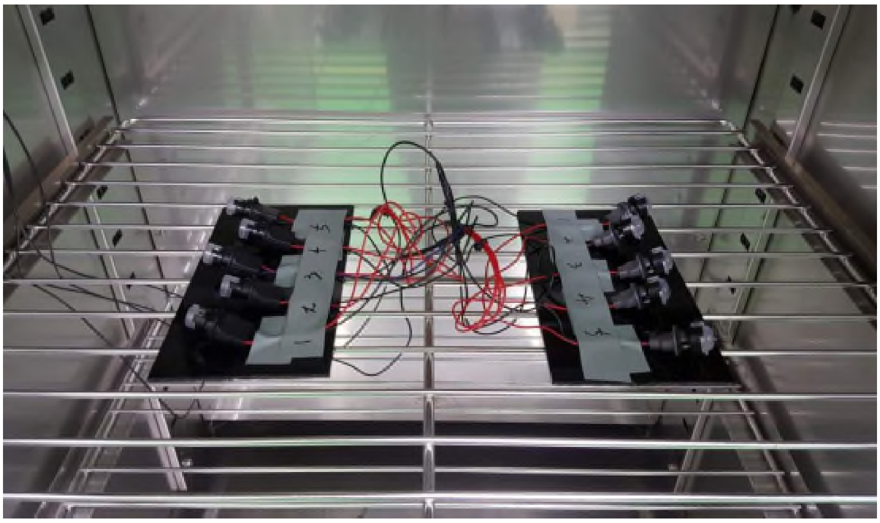
Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
HITACHI	Temp & Humid Chamber	EC-35EXH-R	U5546499	2021.01.13
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22

Remark

IL - QL - 20781	**1-2. Low Temperature Operation**
Test Date	**2020.12.03 ~ 12.06**
Test Sample	**T 6.5 5ea, T 10 5ea**

Test Set-up



After test

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IL - QL - 20781	**1-3. High Temperature Operation**
Test Date	**2020.12.07 ~ 12.10**
Test Sample	**T 6.5 5ea, T 10 5ea**

Test Standard

(1) Put the test sample into the thermostatic chamber at the standard temperature.
Gradually lower the ambient temperature to the minimum dependable operating temperature **(T1, +80℃)**, and leave it untouched at the temperature for 1 h or more. **Operate(12V)** it at the temperature for **72 h**.
(2) Check the test sample for malfunction during the operation.
(3) Take the test sample out of the chamber. If its surface is moist with water droplets, remove them and leave it untouched for 2 h or more under the standard condition.
(4) Evaluate the performance, function and visual inspection.

EUT Conditions

Functional states class
12 V, +80˚C, 72h

EUT Conditions

Test items	Parameters	Result	
		T 6.5	T 10
		5 Samples	5 Samples
High Temperature Operation	- The appearance of parts and components shall not have any trace that may adversely affect performance and function. "Breakage, fractures, cracks, scraping, dimensional changes, corrosion, electrolytic corrosion, discoloration, water intrusion, etc." - As for solder cracking, the cracking rate shall be under 50 % when the largest cracking portion on the fillet is viewed from above. - No trace of smoking and firing.	**PASS**	**PASS**

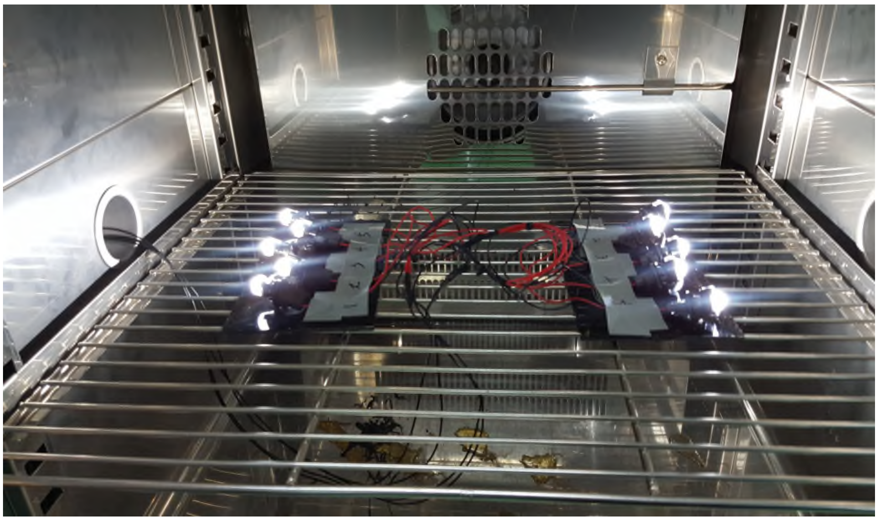
Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
SUKSAN	Hot Dry Chamber	14-02426	IH-G3-199	2021.05.31
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22

Remark

IL - QL - 20781	1-3. High Temperature Operation
Test Date	2020.12.07 ~ 12.10
Test Sample	T 6.5 5ea, T 10 5ea

Test Set-up



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IL - QL - 20781	1-3. High Temperature Operation
Test Date	2020.12.11
Test Sample	T 6.5 5ea, T 10 5ea

After Low Temperature Operation - T 6.5

T 6.5	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	13.5 mV	∞	90.4	PASS
S2	13.4 mV	∞	90.6	PASS
S3	13.6 mV	∞	90.1	PASS
S4	13.7 mV	∞	90.5	PASS
S5	13.1 mV	∞	91.1	PASS

T 6.5	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5272	0.3381	0.3497	PASS
S2	PASS	PASS	5267	0.3382	0.3513	PASS
S3	PASS	PASS	5260	0.3383	0.3496	PASS
S4	PASS	PASS	5252	0.3386	0.3510	PASS
S5	PASS	PASS	5176	0.3405	0.3511	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	**1-3. High Temperature Operation**
Test Date	**2020.12.11**
Test Sample	**T 6.5 5ea, T 10 5ea**

After Low Temperature Operation - T 10

T 10	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	21.2 mV	∞	64.4	PASS
S2	23.1 mV	∞	64.6	PASS
S3	23.7 mV	∞	64.3	PASS
S4	23.5 mV	∞	64.6	PASS
S5	22.8 mV	∞	64.8	PASS

T 10	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5223	0.3393	0.3506	PASS
S2	PASS	PASS	5225	0.3393	0.3501	PASS
S3	PASS	PASS	5236	0.3390	0.3502	PASS
S4	PASS	PASS	5211	0.3397	0.3512	PASS
S5	PASS	PASS	5211	0.3396	0.3511	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	1-4. Thermal Cycle
Test Date	2020.12.11 ~ 12.22
Test Sample	T 6.5 5ea, T 10 5ea

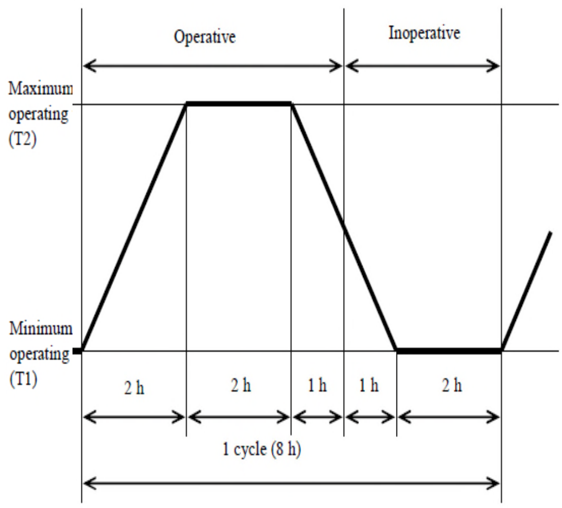
Test Standard

(1) Put the test sample into the thermostatic chamber at the standard temperature. Gradually lower the ambient temp to the minimum dependable operating temp **(T1, -40℃)**, and leave it untouched at the temperature for 1h. Repeat the pattern specified in Fig 4. **30 cycles**, and operate it for the period specified in Fig. 4.

(2) Check the test sample for malfunction during the operation.

(3) After completion of 30 cycles of the test, take the test sample out of the chamber, and leave it untouched under the standard condition for 2 h or more.

(4) Evaluate the performance, function and visual inspection.



EUT Conditions

Functional states class
12 V, Tmax 80˚C, Tmin -40˚C

EUT Conditions

Test items	Parameters	Result	
		T 6.5	T 10
		5 Samples	5 Samples
Thermal Cycle	- No breakage, fractures, cracks, scraping, dimensional changes, corrosion, electrolytic corrosion, discoloration, water intrusion, etc.” - Cracking rate shall be under 50 % - No trace of smoking and firing.	PASS	PASS

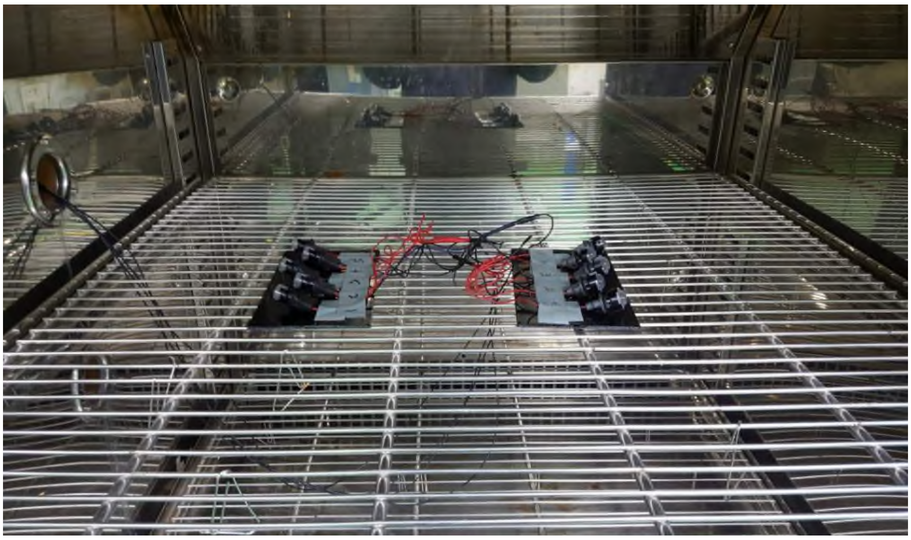
Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EMS	Temp & Humid Chamber	ECTH800	IH-G3-125	2021.01.13
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22

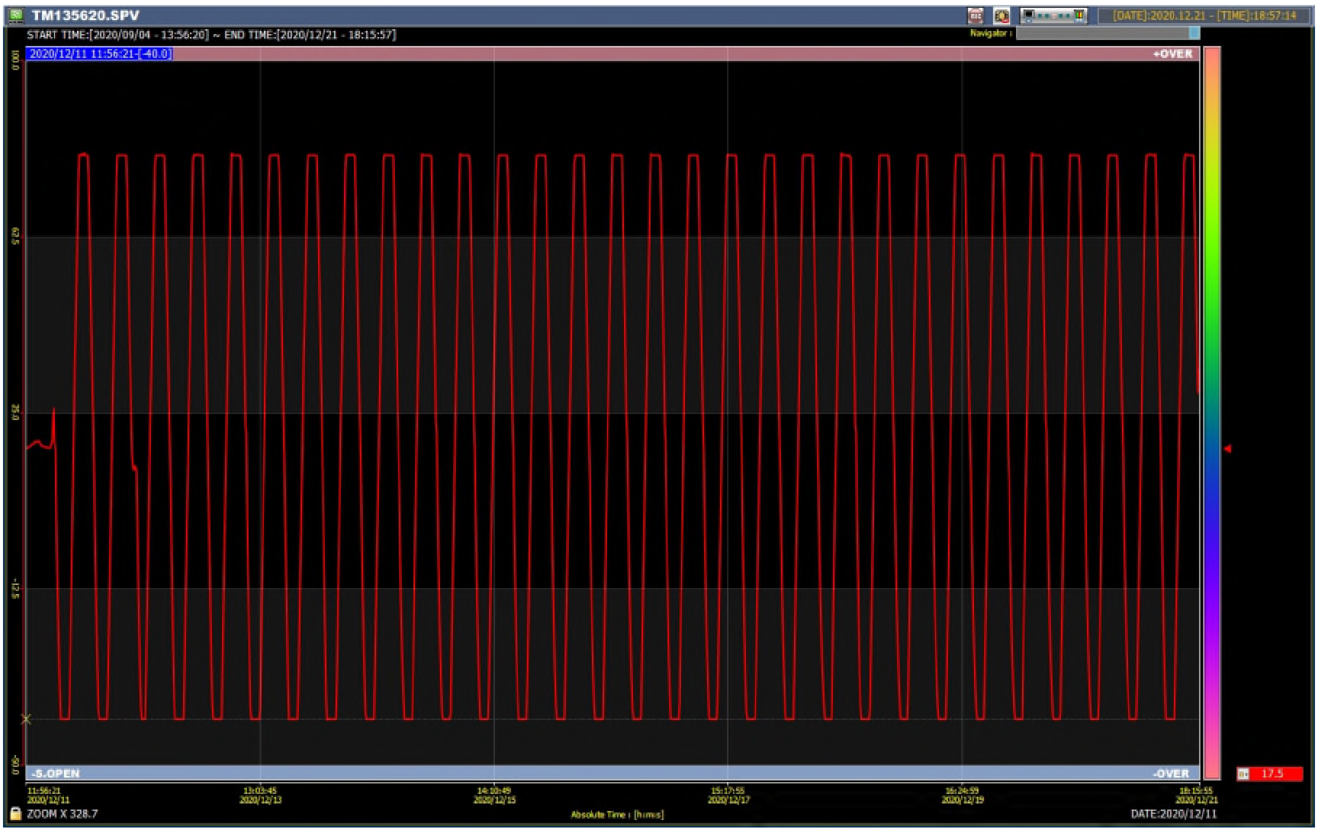
Remark

IL - QL - 20781	1-4. Thermal Cycle
Test Date	2020.12.11 ~ 12.22
Test Sample	T 6.5 5ea, T 10 5ea

Test Set-up

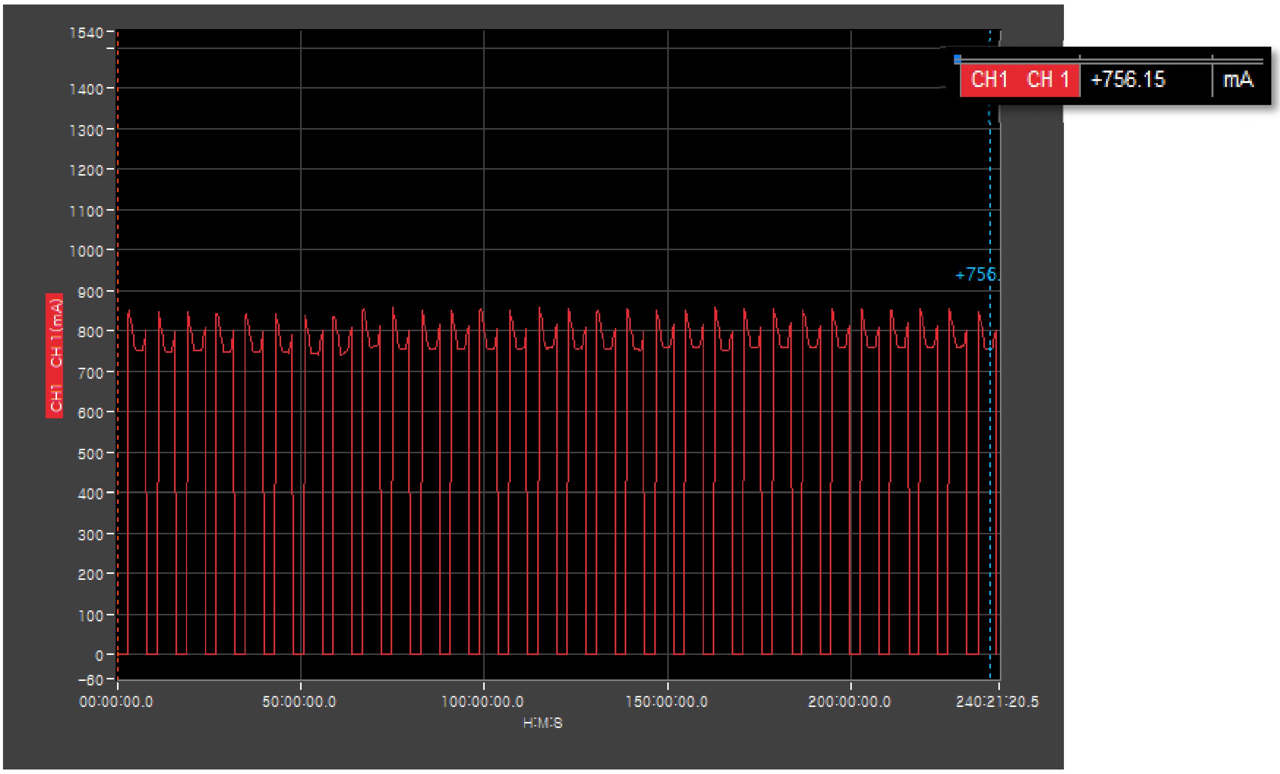


Test Profile - Temperature

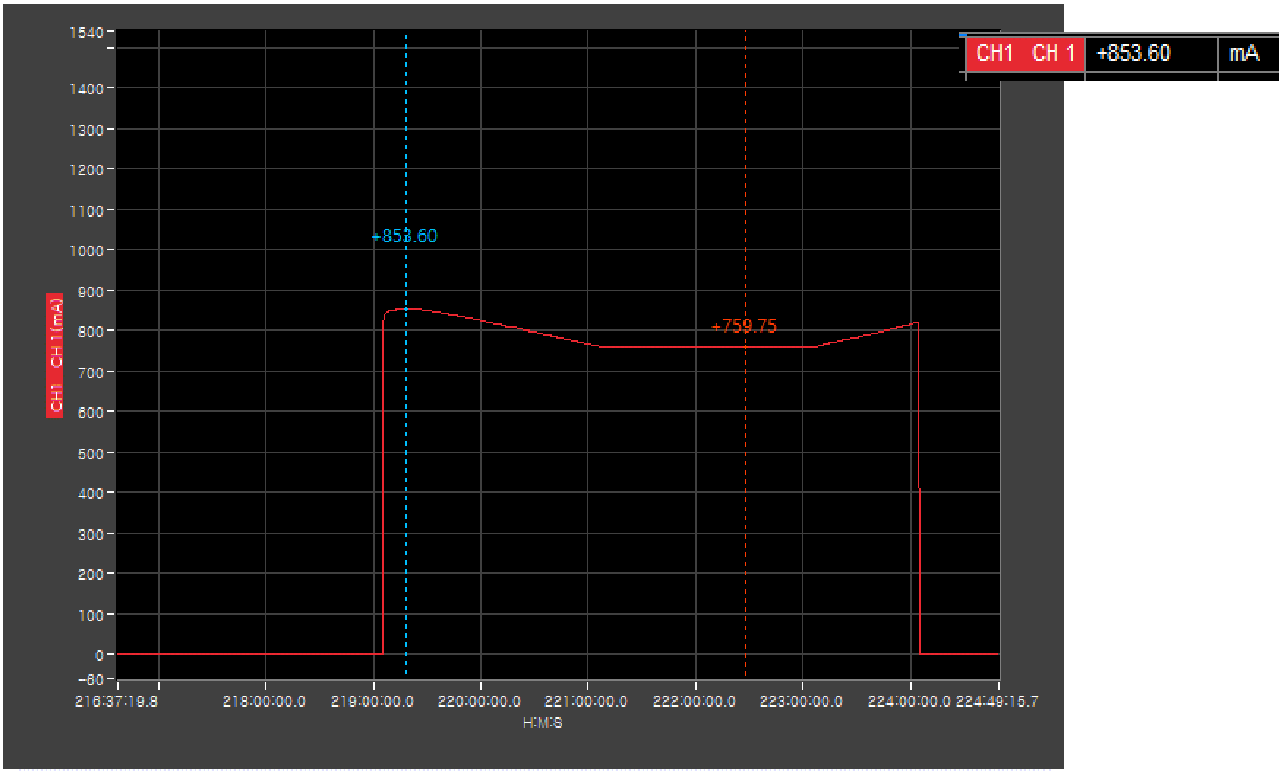


IL - QL - 20781	1-4. Thermal Cycle
Test Date	2020.12.11 ~ 12.22
Test Sample	T 6.5 5ea, T 10 5ea

Test Set-up

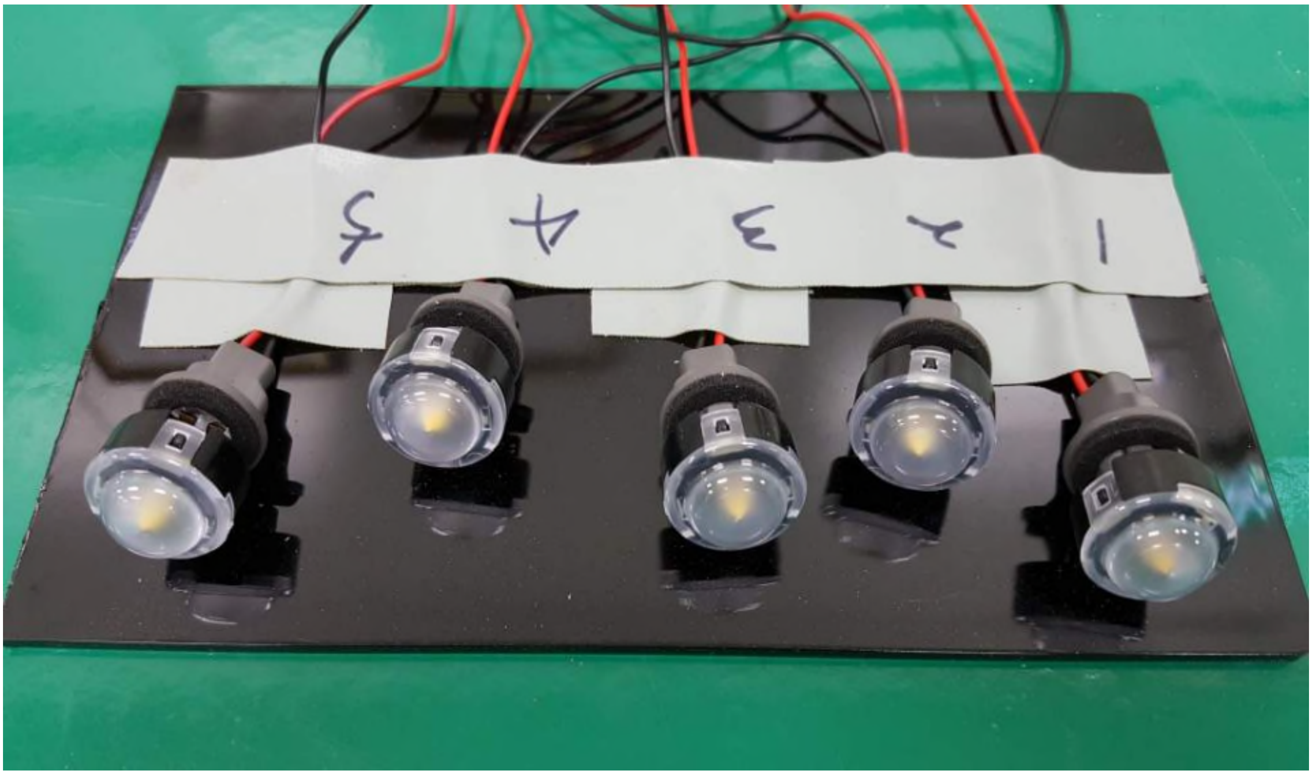


Test Profile - Current

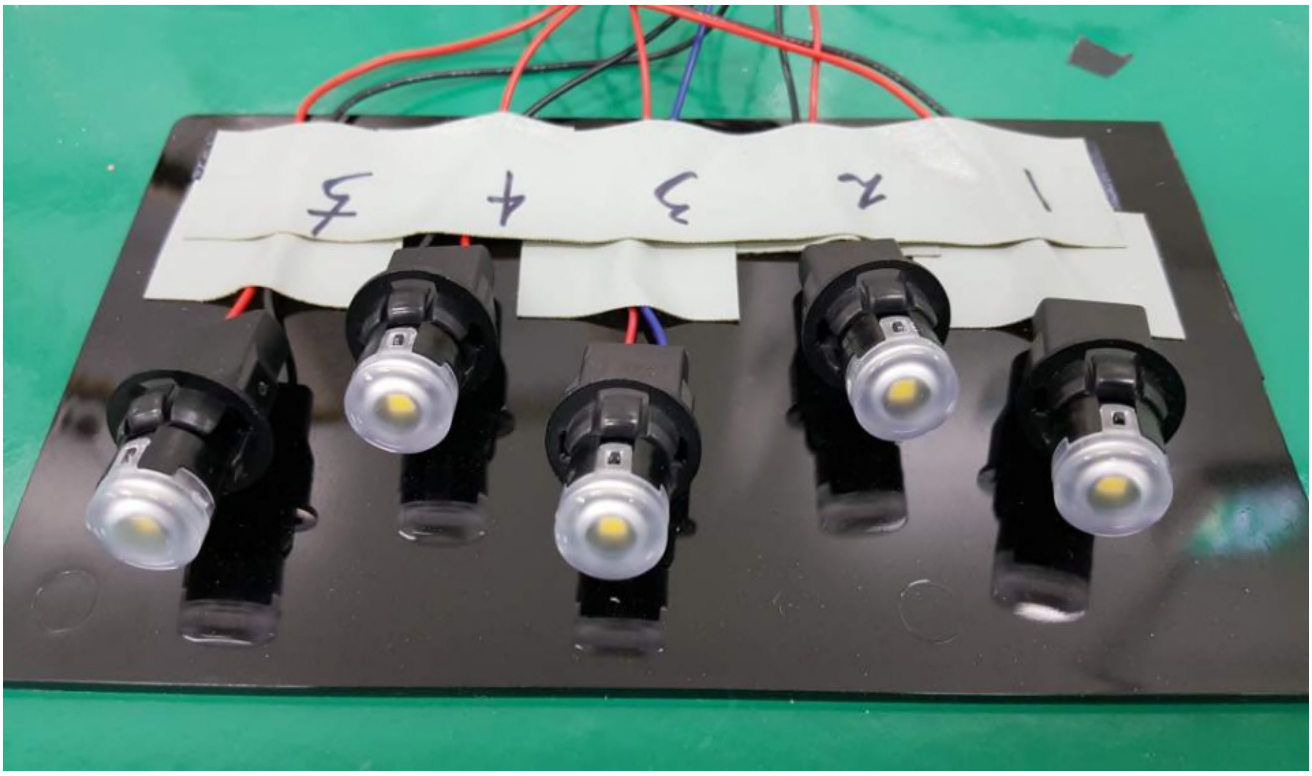


IL - QL - 20781	1-4. Thermal Cycle
Test Date	2020.12.11 ~ 12.22
Test Sample	T 6.5 5ea, T 10 5ea

After test picture - T 6.5



After test picture - T 10



IL - QL - 20781	1-4. Thermal Cycle
Test Date	2020.12.23
Test Sample	T 6.5 5ea, T 10 5ea

After Low Temperature Operation - T 6.5

T 6.5	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	14.0 mV	∞	90.8	PASS
S2	14.0 mV	∞	91.0	PASS
S3	14.0 mV	∞	90.5	PASS
S4	13.8 mV	∞	90.8	PASS
S5	14.0 mV	∞	91.4	PASS

T 6.5	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5227K	0.3392	0.3514	PASS
S2	PASS	PASS	5227K	0.3393	0.3527	PASS
S3	PASS	PASS	5220K	0.3394	0.3510	PASS
S4	PASS	PASS	5209K	0.3398	0.3524	PASS
S5	PASS	PASS	5139K	0.3416	0.3524	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	1-4. Thermal Cycle
Test Date	2020.12.23
Test Sample	T 6.5 5ea, T 10 5ea

After Low Temperature Operation - T 10

T 10	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	**12V**	
S1	23.8 mV	∞	64.7	PASS
S2	23.7 mV	∞	64.9	PASS
S3	23.5 mV	∞	64.6	PASS
S4	23.6 mV	∞	64.9	PASS
S5	23.7 mV	∞	64.7	PASS

T 10	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	**5184K**	0.3404	0.3519	PASS
S2	PASS	PASS	5189K	0.3402	0.3513	PASS
S3	PASS	PASS	5206K	0.3398	0.3514	PASS
S4	PASS	PASS	5179K	0.3406	0.3524	PASS
S5	PASS	PASS	5184K	0.3404	0.3518	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	**1-5. Thermal Shock**
Test Date	**2020.12.24 ~ 12.25**
Test Sample	**T 6.5 5ea, T 10 5ea**

Test Standard

(1) Put the test sample into the thermostatic chamber which has been previously adjusted at the minimum dependable storage temperature **(T5, -40℃)**, and leave it untouched for 2 h or more.
After taking it out of the chamber, immediately put it into the thermostatic chamber which has been adjusted at the maximum dependable storage temperature **(T6, 105℃)**.
After leaving it untouched for the period specified in Table 3, immediately put it into the thermostatic chamber adjusted at the minimum dependable storage temperature (T5), and leave it untouched for the period specified in Table 3.
Taking this procedure as one cycle (see Fig. 5), perform the test 6 cycles continuously.
(2) After completion of **6 cycles** of the test, take the test sample out of the chamber and leave it untouched under the standard condition for 2 h or more.
(3) Evaluate the performance, function and visual inspection. (Mass: under 0.2 - **Exposure time 1h**)

EUT Conditions

Functional states class
Assembly not installed

EUT Conditions

Test items	**Parameters**	**Result**	
		T 6.5	**T 10**
		5 Samples	**5 Samples**
Thermal Shock	- No breakage, fractures, cracks, scraping, dimensional changes, corrosion, electrolytic corrosion, discoloration, water intrusion, etc.” - Cracking rate shall be under 50 % - No trace of smoking and firing.	**PASS**	**PASS**

Test Instruments

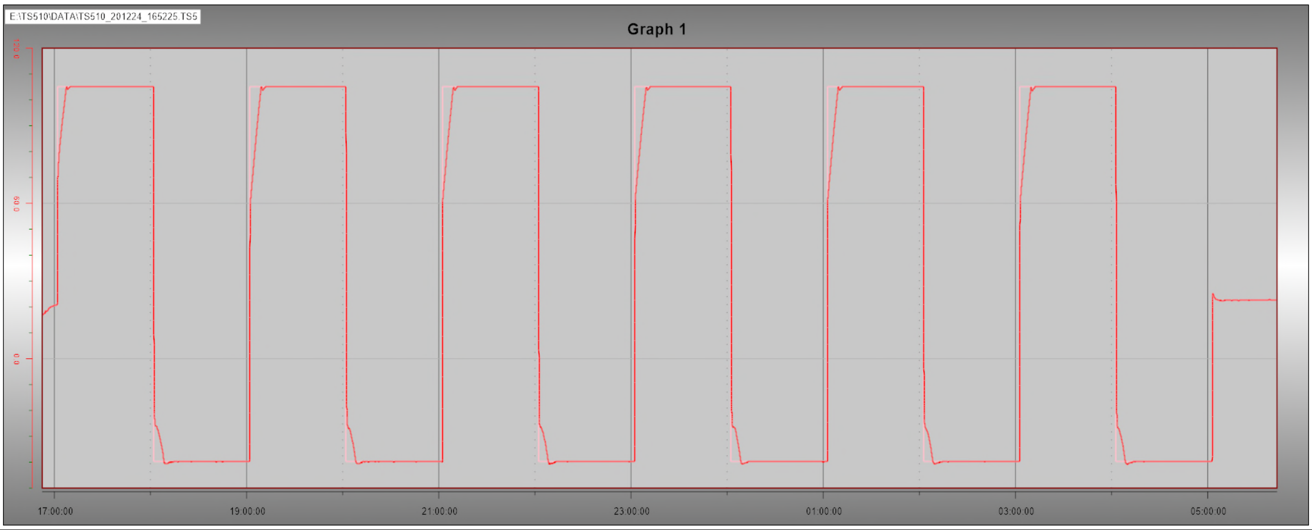
Manufacture	**Test Equipment**	**Model**	**Serial**	**Cal. Due**
EMS	Thermal Shock Tester	ETST-3-150	18071001	2021.01.13
-	-	-	-	-

Remark

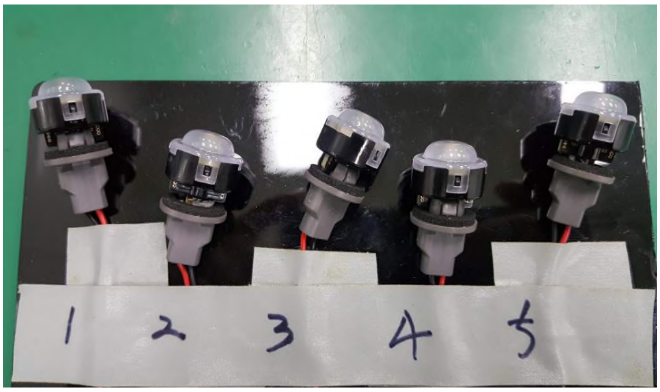
IL - QL - 20781	1-5. Thermal Shock
Test Date	2020.12.24 ~ 12.25
Test Sample	T 6.5 5ea, T 10 5ea

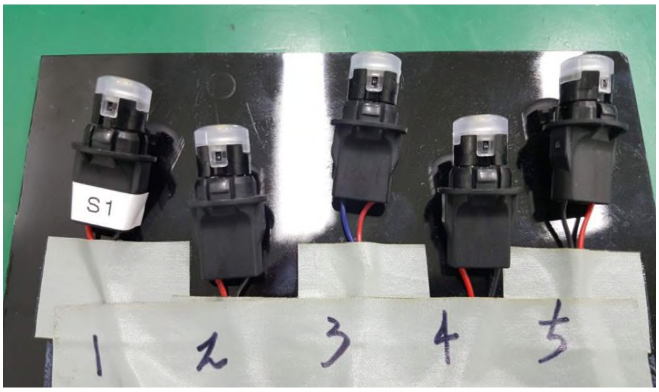
Test Set-up


Test Profile - Temperature



After test picture





IL - QL - 20781	**1-5. Thermal Shock**
Test Date	**2021.01.04**
Test Sample	**T 6.5 5ea, T 10 5ea**

After Low Temperature Operation - T 6.5

T 6.5	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	13.5 mV	∞	91.3	PASS
S2	14.1 mV	∞	91.3	PASS
S3	13.8 mV	∞	90.8	PASS
S4	13.7 mV	∞	91.2	PASS
S5	13.9 mV	∞	91.9	PASS

T 6.5	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5191K	0.3402	0.3524	PASS
S2	PASS	PASS	5196K	0.3402	0.3535	PASS
S3	PASS	PASS	5193K	0.3401	0.3515	PASS
S4	PASS	PASS	5176K	0.3407	0.3533	PASS
S5	PASS	PASS	5108K	0.3424	0.3533	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	**1-5. Thermal Shock**
Test Date	**2021.01.04**
Test Sample	**T 6.5 5ea, T 10 5ea**

After Low Temperature Operation - T 10

T 10	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	23.4 mV	∞	65.1	PASS
S2	24.2 mV	∞	65.4	PASS
S3	24.1 mV	∞	65.0	PASS
S4	24.5 mV	∞	65.3	PASS
S5	24.7 mV	∞	65.2	PASS

T 10	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5159K	0.3410	0.3527	PASS
S2	PASS	PASS	5152K	0.3413	0.3526	PASS
S3	PASS	PASS	5173K	0.3407	0.3524	PASS
S4	PASS	PASS	5149K	0.3414	0.3534	PASS
S5	PASS	PASS	5149K	0.3413	0.3530	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	289	24190030	2021.01.13

IL - QL - 20781	1-6. Temperature Humidity Cycle
Test Date	2021.01.06 ~ 01.11
Test Sample	T 6.5 5ea, T 10 5ea

Test Standard

(1) Put the test sample into a thermo-hygrostat at the room temperature, and adjust the temp and humid inside the thermo-hygrostat to 25˚C and 65%, respectively.
After reaching these conditions, leave the test sample for 2.5 h or more.
Perform the test specified in Fig.7 **5 cycles**.

(2) Check for malfunction during the operation.

(3) After completion of 5 cycles, take the test sample out of the thermo-hygrostat and leave it for 2 h or more in the standard condition.

(4) Evaluate the performance, function and visual inspection.

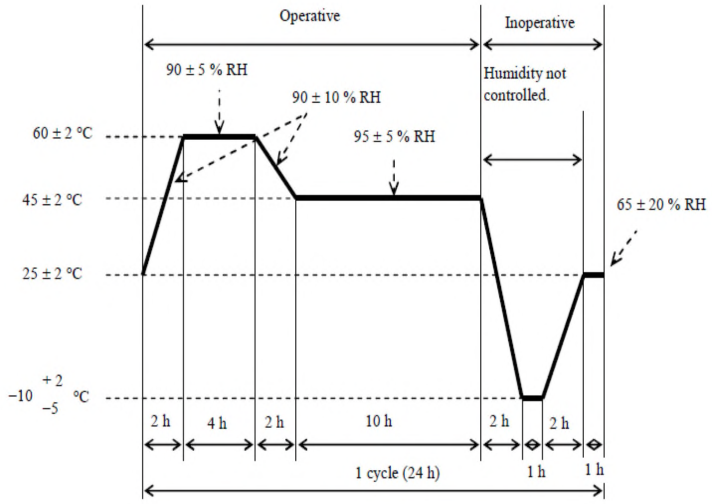


Fig. 7 Temperature – Humidity Cycle Test Pattern

EUT Conditions

Functional states class
12 V, Tmax 60˚C, Tmin -10˚C x 65~95%, 1 Cycle (24h)

EUT Conditions

Test items	Parameters	Result	
		T 6.5	T 10
		5 Samples	5 Samples
Temperature Humidity Cycle	- No breakage, fractures, cracks, scraping, dimensional changes, corrosion, electrolytic corrosion, discoloration, water intrusion, etc.” - Cracking rate shall be under 50 % - No trace of smoking and firing.	PASS	PASS

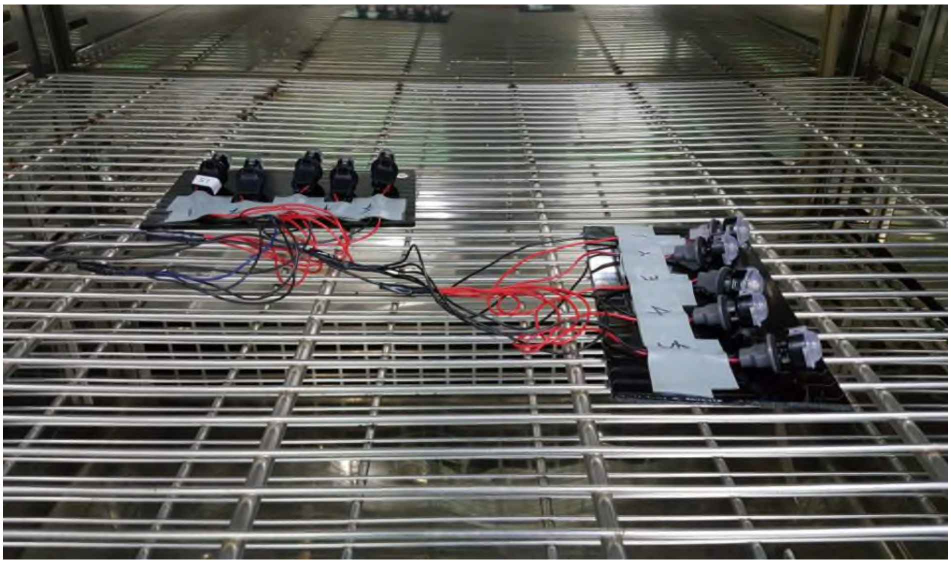
Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EMS	Temp & Humid Chamber	ECIR-1300A	15010701	2021.05.31
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22

Remark

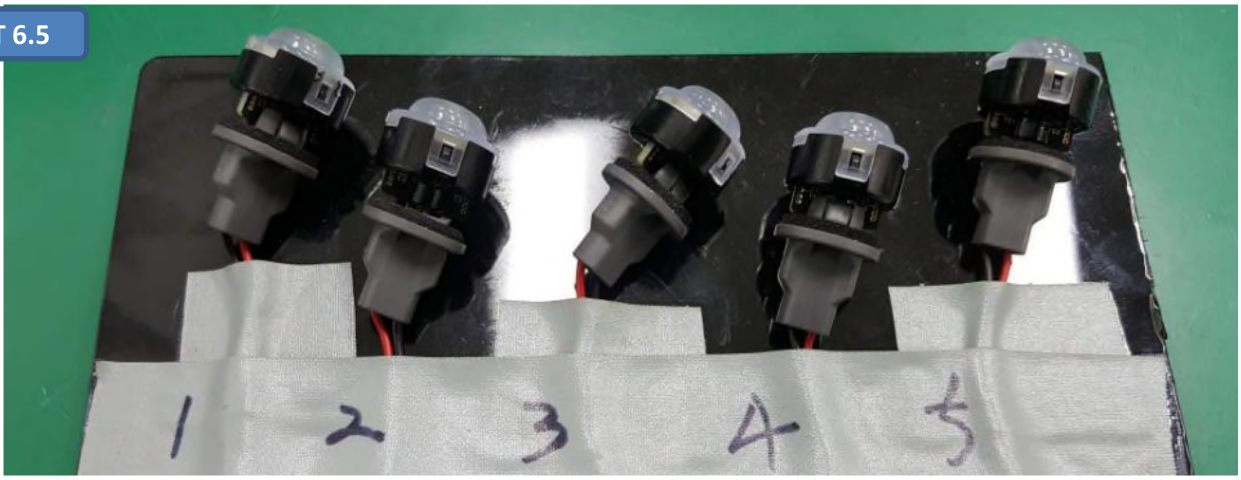
IL - QL - 20781	**1-6. Temperature Humidity Cycle**
Test Date	**2021.01.06 ~ 01.11**
Test Sample	**T 6.5 5ea, T 10 5ea**

Test Set-up

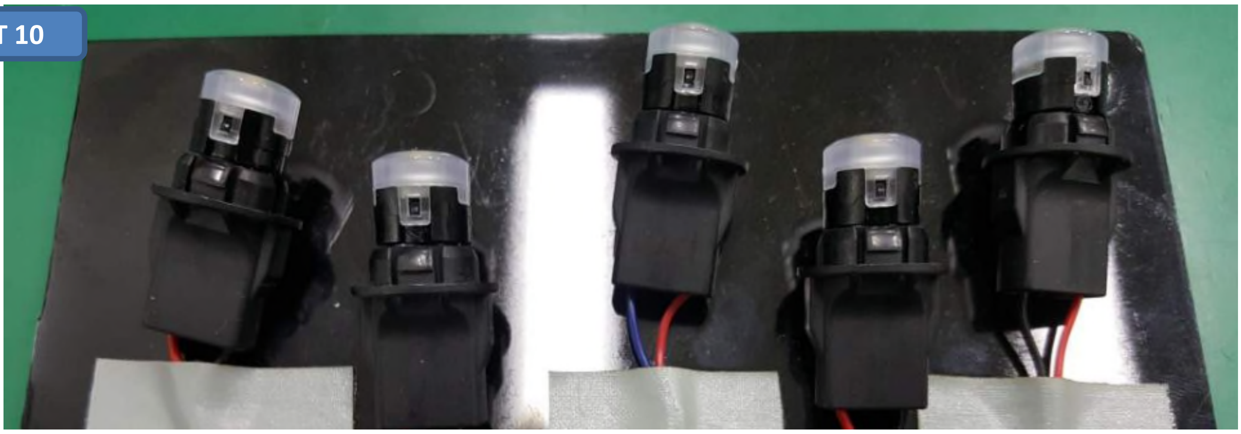


Test Profile - Current

T 6.5



T 10



IL - QL - 20781	1-7. Performance evaluation/Functional evaluation, Final
Test Date	2021.01.11
Test Sample	T 6.5 5ea, T 10 5ea

Performance & Functional evaluation for T 6.5

T 6.5	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	14.0 mV	∞	91.2	PASS
S2	14.0 mV	∞	91.3	PASS
S3	13.9 mV	∞	90.7	PASS
S4	14.1 mV	∞	91.3	PASS
S5	13.8 mV	∞	91.8	PASS

T 6.5	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5191K	0.3403	0.3525	PASS
S2	PASS	PASS	5196K	0.3404	0.3534	PASS
S3	PASS	PASS	5193K	0.3402	0.3513	PASS
S4	PASS	PASS	5176K	0.3405	0.3535	PASS
S5	PASS	PASS	5108K	0.3425	0.3534	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	189	93280095	2021.05.31

IL - QL - 20781	**1-7. Performance evaluation/Functional evaluation, Final**
Test Date	**2021.01.11**
Test Sample	**T 6.5 5ea, T 10 5ea**

Performance & Functional evaluation for T 10

T 10	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	25.0 mV	∞	65.1	PASS
S2	24.9 mV	∞	65.4	PASS
S3	24.5 mV	∞	65.1	PASS
S4	24.7 mV	∞	65.4	PASS
S5	24.3 mV	∞	65.3	PASS

T 10	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5159K	0.3411	0.3526	PASS
S2	PASS	PASS	5152K	0.3414	0.3525	PASS
S3	PASS	PASS	5173K	0.3406	0.3526	PASS
S4	PASS	PASS	5149K	0.3412	0.3536	PASS
S5	PASS	PASS	5149K	0.3414	0.3531	PASS

Test Instruments

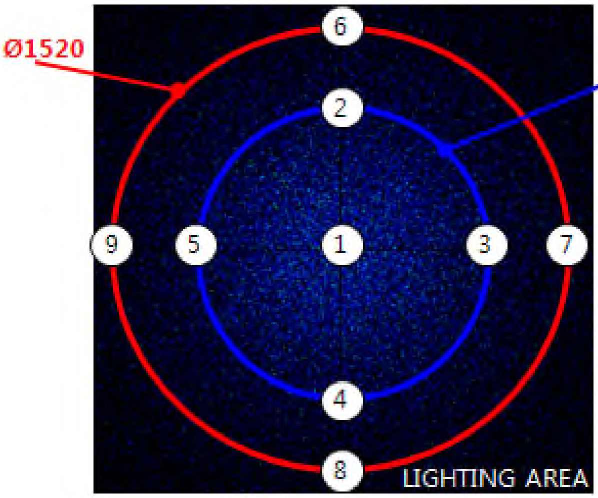
Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	189	93280095	2021.05.31

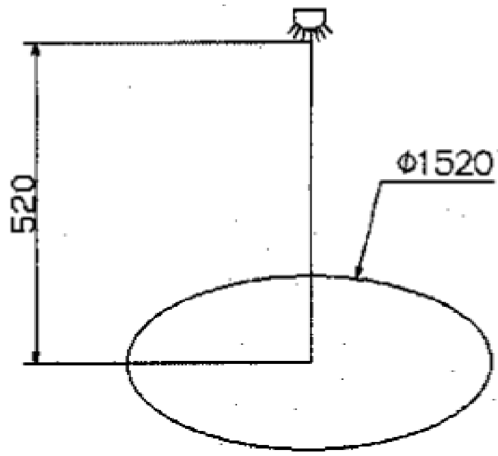
IL - QL - 20781	2. Illuminnation - T 6.5
Test Date	2021.01.08
Test Sample	T 6.5 5ea

Test Standard

More than 5Lx in this area. Center illumication is less than 180 lx

Test Standard





EUT Conditions

Operating class
12V

EUT Conditions

Test items	Parameters	Result
Illuminnation - T 6.5	More than 5Lx in this area. Center illumication is less than 180Lx	T 6.5
		5 Samples
		PASS

Illuninance

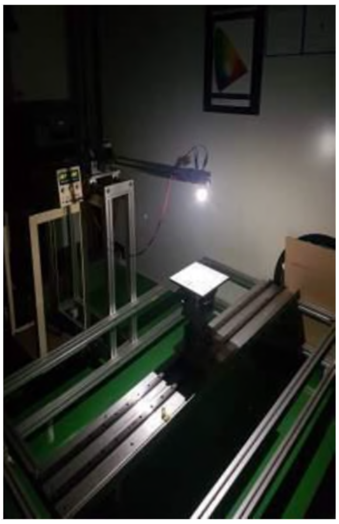
T 6.5		Illuninance									Test result
		P1 (CTR)	P2	P3	P4	P5	P6	P7	P8	P9	
> 5 lx Center : < 180 lx	S1	24.4	21.8	21.9	22.0	22.9	11.4	12.2	11.8	11.0	PASS
	S2	25.2	22.1	22.0	22.0	22.8	11.7	11.8	11.9	11.1	PASS
	S3	24.6	22.3	22.1	21.9	22.8	11.2	11.8	12.0	11.0	PASS
	S4	24.7	22.0	21.8	21.7	23.0	11.6	11.9	12.0	10.9	PASS
	S5	24.9	22.3	22.0	22.1	22.9	11.1	12.1	11.9	10.8	PASS

IL - QL - 20781	2. Illuminnation - T 10
Test Date	2021.01.08
Test Sample	T 10 5ea

Test Standard

Measuring position : 200㎜ / MIN 340lx

Test Standard



Measuring position : Height 200㎜ / Min 340lx

EUT Conditions

Operating class
12V

EUT Conditions

Test items	Parameters	Result
Illuminnation - T 10	Min 340lx	T 10
		5 Samples
		PASS

Illuninance

T 10	Illuninance					Test result
	S1	S2	S3	S4	S5	
Min 340lx	361.0	359.8	354.3	357.2	358.1	PASS
						PASS
						PASS
						PASS
						PASS

IL - QL - 20781	**3. High temp & humidity actual vehicle test**
Test Date	**2021.01.06 ~ 01.11**
Test Sample	**T 6.5 5ea, T 10 5ea**

Test Standard

(1) **50℃x 50% x 11h, LAMP On / 50℃x 50% x 1h, LAMP Off**
TOTAL : 10 Cycle

EUT Conditions

Functional states class
50℃x 50% x 11h, LAMP On / 50℃x 50% x 1h, LAMP Off

EUT Conditions

Test items	Parameters	Result	
		T 6.5	T 10
		5 Samples	5 Samples
High temp & humidity actual vehicle test	- The appearance of parts and components shall not have any trace that may adversely affect performance and function. "Breakage, fractures, cracks, scraping, dimensional changes, corrosion, electrolytic corrosion, discoloration, water intrusion, etc." - As for solder cracking, the cracking rate shall be under 50 % when the largest cracking portion on the fillet is viewed from above. - No trace of smoking and firing.	PASS	PASS

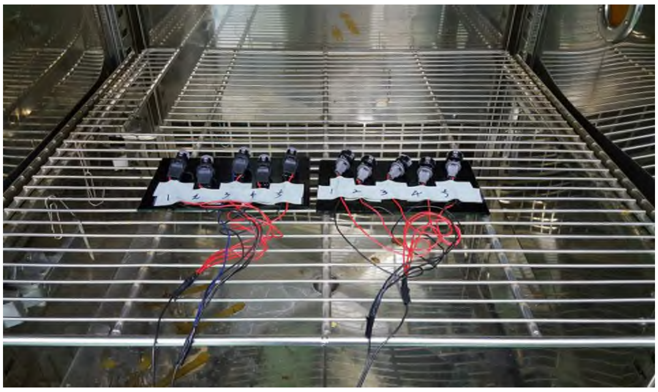
Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EMS	ECTH-300A-2	G-0210NP	13082001-2	2021.01.13
KAPJIN	DC POWER SUPPLY	KJPP-3030	06101264	2021.05.31
-	-	-	-	-

Remark

IL - QL - 20781	3. High temp & humidity actual vehicle test
Test Date	2021.01.06 ~ 01.11
Test Sample	T 6.5 5ea, T 10 5ea

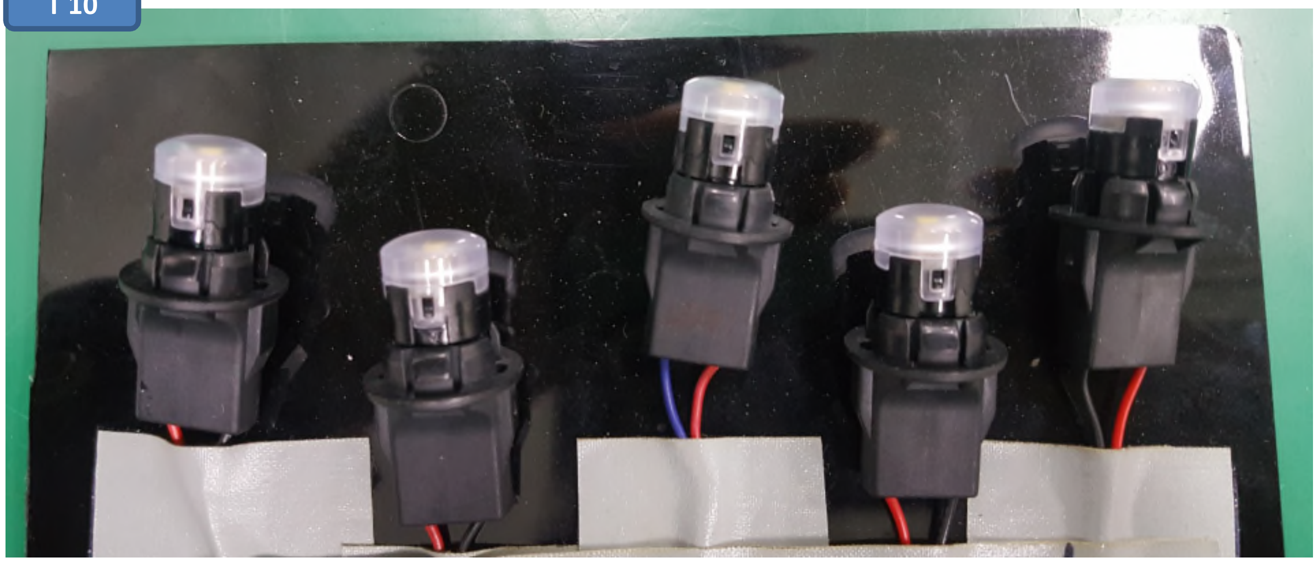
Test Set-up



Test Profile - Current

T 6.5

T 10



IL - QL - 20781	**3. High temp & humidity actual vehicle test**
Test Date	**2021.01.11**
Test Sample	**T 6.5 5ea, T 10 5ea**

After Low Temperature Operation - T 6.5

T 6.5	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	**12V**	
S1	13.8 mV	∞	90.4	PASS
S2	13.9 mV	∞	90.6	PASS
S3	14.1 mV	∞	90.1	PASS
S4	13.5 mV	∞	90.5	PASS
S5	13.7 mV	∞	91.1	PASS

T 6.5	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5208	0.3394	0.3506	PASS
S2	PASS	PASS	5202	0.3392	0.3518	PASS
S3	PASS	PASS	5204	0.3394	0.3505	PASS
S4	PASS	PASS	5202	0.3393	0.3516	PASS
S5	PASS	PASS	5201	0.3402	0.3515	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	189	93280095	2021.05.31

IL - QL - 20781	**3. High temp & humidity actual vehicle test**
Test Date	**2021.01.11**
Test Sample	**T 6.5 5ea, T 10 5ea**

After Low Temperature Operation - T 10

T 10	Functional Test			Test result
	Voltage Drop	Insulation Resistance	Current Check 23˚C	
SPECIFICATION	0.1V (MAX)	Shall be 1MΩ min.	12V	
S1	23.7 mV	∞	65.3	PASS
S2	24.1 mV	∞	65.2	PASS
S3	23.2 mV	∞	65.3	PASS
S4	23.6 mV	∞	65.1	PASS
S5	23.8 mV	∞	65.5	PASS

T 10	Performance Test					Test result
	LED ON	Flicker & Brightness	LED color temperature	LED COLOR		
SPECIFICATION	5 Performance Evaluation Points.		WHITE: 5000K	X 0.3366~0.3551	Y 0.3369~0.3760	
S1	PASS	PASS	5185	0.3404	0.3510	PASS
S2	PASS	PASS	5185	0.3405	0.3507	PASS
S3	PASS	PASS	5188	0.3404	0.3505	PASS
S4	PASS	PASS	5182	0.3405	0.3511	PASS
S5	PASS	PASS	5181	0.3403	0.3502	PASS

Test Instruments

Manufacture	Test Equipment	Model	Serial	Cal. Due
EVERFINE	Intergrating Sphere	CAS-200	G121958C J1341157D	2021.02.06
GW INSTEK	Power Supply	PSP-2010	E0150560	2021.05.22
FLUKE	RMS Multimeter	189	93280095	2021.05.31